

## Description and choice data for the domain “Travel”

### Description of the choice domain 15, Travel

The prompt question and the universe of five response options in the choice domain **Travel** are as follows. The labels *a*, *b*, *c*, *d* and *e* were not displayed during the experiment and are indicated here to allow cross-referencing with data tables and visualizations below and results in the paper.

The source is Tripadvisor. These are the top five travel destinations, according to the results of an on-line contest where visitors to a Tripadvisor site could make pairwise choices between travel destinations.

Which one of the following travel destinations would you most like to visit?

1. Marrakech, Morocco 
2. Istanbul, Turkey 
3. Hanoi, Vietnam 
4. Siem Reap, Cambodia 
5. Prague, Czech Republic 

The following figure is a screenshot from the actual experiment, with one of the 26 possible menus for this

domain.

**Travel**

Which one of the following travel destinations would you most like to visit?



Istanbul, TurkeyMarrakech, Morocco

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Figure 1: Screenshot for domain Travel

| Menu $A$            | Choice counts |          |          |          |          | Choice proportions |                |                |                |                |
|---------------------|---------------|----------|----------|----------|----------|--------------------|----------------|----------------|----------------|----------------|
|                     | $N_A(a)$      | $N_A(b)$ | $N_A(c)$ | $N_A(d)$ | $N_A(e)$ | $\hat{P}_A(a)$     | $\hat{P}_A(b)$ | $\hat{P}_A(c)$ | $\hat{P}_A(d)$ | $\hat{P}_A(e)$ |
| $\{a, b\}$          | 27            | 13       | -        | -        | -        | 0.675              | 0.325          | -              | -              | -              |
| $\{a, c\}$          | 29            | -        | 11       | -        | -        | 0.725              | -              | 0.275          | -              | -              |
| $\{b, c\}$          | -             | 20       | 20       | -        | -        | -                  | 0.500          | 0.500          | -              | -              |
| $\{a, b, c\}$       | 19            | 10       | 11       | -        | -        | 0.475              | 0.250          | 0.275          | -              | -              |
| $\{a, d\}$          | 30            | -        | -        | 10       | -        | 0.750              | -              | -              | 0.250          | -              |
| $\{b, d\}$          | -             | 24       | -        | 16       | -        | -                  | 0.600          | -              | 0.400          | -              |
| $\{a, b, d\}$       | 20            | 14       | -        | 6        | -        | 0.500              | 0.350          | -              | 0.150          | -              |
| $\{c, d\}$          | -             | -        | 30       | 10       | -        | -                  | -              | 0.750          | 0.250          | -              |
| $\{a, c, d\}$       | 24            | -        | 12       | 4        | -        | 0.600              | -              | 0.300          | 0.100          | -              |
| $\{b, c, d\}$       | -             | 16       | 17       | 7        | -        | -                  | 0.400          | 0.425          | 0.175          | -              |
| $\{a, b, c, d\}$    | 15            | 7        | 10       | 8        | -        | 0.375              | 0.175          | 0.250          | 0.200          | -              |
| $\{a, e\}$          | 13            | -        | -        | -        | 28       | 0.317              | -              | -              | -              | 0.683          |
| $\{b, e\}$          | -             | 9        | -        | -        | 32       | -                  | 0.220          | -              | -              | 0.780          |
| $\{a, b, e\}$       | 11            | 5        | -        | -        | 24       | 0.275              | 0.125          | -              | -              | 0.600          |
| $\{c, e\}$          | -             | -        | 10       | -        | 30       | -                  | -              | 0.250          | -              | 0.750          |
| $\{a, c, e\}$       | 15            | -        | 8        | -        | 17       | 0.375              | -              | 0.200          | -              | 0.425          |
| $\{b, c, e\}$       | -             | 7        | 4        | -        | 29       | -                  | 0.175          | 0.100          | -              | 0.725          |
| $\{a, b, c, e\}$    | 13            | 2        | 2        | -        | 23       | 0.325              | 0.050          | 0.050          | -              | 0.575          |
| $\{d, e\}$          | -             | -        | -        | 8        | 32       | -                  | -              | -              | 0.200          | 0.800          |
| $\{a, d, e\}$       | 13            | -        | -        | 3        | 24       | 0.325              | -              | -              | 0.075          | 0.600          |
| $\{b, d, e\}$       | -             | 9        | -        | 4        | 27       | -                  | 0.225          | -              | 0.100          | 0.675          |
| $\{a, b, d, e\}$    | 8             | 6        | -        | 8        | 18       | 0.200              | 0.150          | -              | 0.200          | 0.450          |
| $\{c, d, e\}$       | -             | -        | 8        | 5        | 27       | -                  | -              | 0.200          | 0.125          | 0.675          |
| $\{a, c, d, e\}$    | 10            | -        | 3        | 2        | 25       | 0.250              | -              | 0.075          | 0.050          | 0.625          |
| $\{b, c, d, e\}$    | -             | 5        | 9        | 2        | 24       | -                  | 0.125          | 0.225          | 0.050          | 0.600          |
| $\{a, b, c, d, e\}$ | 10            | 4        | 4        | 1        | 21       | 0.250              | 0.100          | 0.100          | 0.025          | 0.525          |

Table 1: Observed choice counts and proportions.

## Choice data for domain 15, Travel

Table 1 shows choice counts and choice proportions for this choice domain. For each menu  $A$  and each object  $x \in \{a, b, c, d, e\}$ ,  $N_A(x)$  is the number of participants who chose object  $x$  from menu  $A$  and  $\hat{P}_A(x)$  is the corresponding proportion of participants who chose  $x$  from  $A$ . When  $x \notin A$ , a dash is displayed.

The following figure displays choice proportions for all doubleton and tripleton menus in Barycentric coordinates. See a full description of this graphical representation in the paper. Each panel shows choice proportions for all doubleton and tripleton menus of a different tripleton subset of  $\{a, b, c, d, e\}$ . The downward-pointed (blue) triangle shows the set of ternary choice proportions that are compatible with regularity and the three binary choice proportions, on the corresponding tripleton. The upward-pointed (red) triangle shows the set of ternary choice proportions compatible with the multiplicative inequality and the three binary choice proportions.

```
## Warning in tritrafo(x, y, z): negative components
## Warning in tritrafo(x, y, z): negative components
## Warning in tritrafo(x, y, z): negative components
```

